AMERICAN UNIVERSITY OF BEIRUT

Module 3 Additional Readings

AI Assisted Coding

How You Should Use This Resource

- Read the Core readings after the lecture. These are the minimum supporting readings.
- Use the Practical references while building and debugging the Task Tracker Kanban board. They are lookup references, not assigned essays.
- Use Optional deeper dives only after the core work is done. They are not required for the quiz, deliverables, or grading.

Core Readings: Students Should Read These

Recommended time: about 50-70 minutes total if students focus only on the listed sections. These readings directly support the Module 3 slides, demos, and deliverables.

Core 1. GitHub Docs - Best practices for using GitHub Copilot

Link: GitHub Docs - Best practices for using GitHub Copilot

Purpose: Use this as the main AI-assisted coding mindset reading for Module 3. It supports the lecture loop: ask, inspect, run, test, and refine.

Read / use this part: Focus on creating thoughtful prompts, breaking down complex tasks, providing examples, checking Copilot's work, and using tests/tooling to validate suggestions.

Core 2. GitHub Docs - Asking GitHub Copilot questions in your IDE

Link: GitHub Docs - Asking GitHub Copilot questions in your IDE

Purpose: Supports Part 3.1: open the real Task Tracker repo, ask Copilot questions from inside the IDE, and verify that it is reading the right code.

Read / use this part: Skim the sections on IDE chat, smart actions, and adding context. Pay attention to workflows where you ask about selected code or files instead of asking vague repo-wide questions.

Core 3. VS Code Docs - Inline chat

Link: VS Code Docs - Inline chat

Purpose: Supports Parts 3.3 and 3.4: use inline chat on a selected handler, modal section, render function, or refactor target instead of asking for a full-file rewrite.

Read / use this part: Read the editor inline chat section, especially selecting a block of code to scope the prompt and reviewing the inline diff with Keep/Undo.

Core 4. MDN - HTML Drag and Drop API

Link: MDN - HTML Drag and Drop API

Purpose: Supports Part 3.2: the Kanban board uses native browser drag-and-drop, not a drag library.

Read / use this part: Focus on the concepts and basic steps: draggable item, transferred data, drop target, dragstart, dragover, and drop. Skip file-upload examples.

Core 5. MDN - Using the Fetch API

Link: MDN - Using the Fetch API

Purpose: Supports fetchTasks, renderBoard, POST/PATCH form submission, and error-state handling in the frontend.

Read / use this part: Focus on sending JSON, reading JSON responses, and checking response status. The most important detail for Module 3: fetch does not reject just because the server returns 404 or 422; your code must check status or ok.

Core 6. GitHub Docs - Writing tests with GitHub Copilot

Link: GitHub Docs - Writing tests with GitHub Copilot

Purpose: Supports Part 3.5: use Copilot to brainstorm and draft tests, then verify the tests yourself.

Read / use this part: Read the introduction and the Copilot Chat testing workflow. Notice the need for specific test requirements and verification after generation.

Practical References: Use These While Working

These are not required sit-and-read assignments. Keep them open while completing the hands-on work, checking generated code, diagnosing browser/API issues, or writing tests.

Practical 1. FastAPI - Testing

Link: FastAPI - Testing

Purpose: Use while adding backend tests for PATCH edge cases and status-transition behavior.

Read / use this part: Focus on TestClient, naming test functions with test_, and using the client like httpx. Skip advanced deployment or async topics not used in Module 3.

Practical 2. pytest - Documentation

Link: pytest - Documentation

Purpose: Use as the reference for running pytest, reading failures, and understanding small readable test functions.

Read / use this part: Use it as a lookup reference. For Module 3, you mainly need naming conventions, running selected tests, and interpreting failure output.

Student note: Do not drift into fixtures, plugins, or large testing architecture unless your instructor assigns it later.

Practical 3. MDN - DataTransfer

Link: MDN - DataTransfer

Purpose: Use while checking dragstart/drop code that stores the task id with dataTransfer and sets move behavior.

Read / use this part: Focus on setData/getData, effectAllowed, and dropEffect. Skip clipboard and file-transfer sections.

Student note: This is more specific than the main Drag and Drop overview and is useful when Copilot generates confusing dataTransfer logic.

Practical 4. VS Code Docs - Manage context for AI

Link: VS Code Docs - Manage context for AI

Purpose: Use when Copilot answers from generic patterns instead of the actual Task Tracker files.

Read / use this part: Focus on adding files, folders, symbols, terminal output, and codebase context to chat. Use this to make prompts grounded in the files students are actually editing.

Optional Deeper Dives: For Students Who Want More Detail

These readings are useful, but they go beyond what students need to complete Module 3. Use them selectively after the core and practical references.

Optional 1. GitHub Docs - Prompt engineering for GitHub Copilot Chat

Link: GitHub Docs - Prompt engineering for GitHub Copilot Chat

Purpose: Use for extra practice writing prompts that include context, constraints, examples, and output expectations.

Read / use this part: Read selectively. The core Module 3 prompts already model the pattern; this is for students who want more general Copilot prompting guidance.

Optional 2. Martin Fowler / Thoughtworks - Exploring Generative AI

Link: Martin Fowler - Exploring Generative AI

Purpose: Use for broader practitioner context on AI-assisted software delivery after students finish the module.

Read / use this part: Read selectively. Choose posts about human oversight, AI-assisted delivery practice, or code review. Do not treat the whole series as assigned reading.

Student note: This is intentionally optional because Module 3 is a hands-on AI Assisted Coding module, not a general software engineering theory module.